Prediction models for survival analysis

M2 Données massives en santé

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Introduction

Prediction models

There are thousands of prediction models in the medical literature

Use "baseline" variables to predict an outcome (most of time occurrence of an event): diagnosis or prognosis

- Binary (dead/alive): logistic regression
- Survival: Cox regression
- Continuous (rare): linear regression

But any other approach could be used (e.g. random forests, neural networks, SVM, etc.)

Examples

Short-term outcome (binary)

- Hospital mortality – SAPS-II (Simplified Acute Physiology Score), APACHE II/III

Longer-term outcome

- 10-year cardiovascular disease risk – Framingham risk score, QRISK2
- 2-year non-relapse mortality – HCT-CI (comorbidity index)
- 12-year recurrence after radical prostatectomy

What is a good prediction model?

"All models are wrong, but some are useful" Box (1976)

- **Multiple definitions and perspectives:**
  - A model is *"good"* if it is **useful** – but how do we define usefulness?
  - The ultimate goal: positively impacting **patient outcomes** (gold standard, but rare).
- **Steps before assessing therapeutic or clinical impact:**
  - Focus first on evaluating how well the model **performs**.
- **Key Aspects of model performance:**
  1. **Calibration:** How closely do predicted probabilities align with observed outcomes?
  2. **Discrimination:** How effectively does the model differentiate between cases (e.g., diseased) and controls (e.g., non-diseased)?

Evaluating performance

Concordance Index for Survival Analysis

What is the Concordance Index?

- The most common method for evaluating survival models is based on the relative risk of an event rather than the absolute survival times.
- This is done by calculating the concordance probability or the concordance index (C-index).

$$\mathbb{C} = \mathbb{P}(\eta_i > \eta_j | T_i < T_j), \tag{1}$$

where η_i is the risk score for individual i and T_i is the observed survival time.

Key References: Harrell (1982), Uno (2011), Gerds (2013)

C-index by Harrell (1982)

$$\hat{\mathbb{C}}_{\mathsf{H}} = \frac{\sum_{i \neq j} 1(\hat{\eta}_i > \hat{\eta}_j) 1(T_i < T_j, \delta_i = 1)}{\sum_{i \neq j} 1\left(T_i < T_j, \delta_i = 1\right)}, \tag{2}$$

where $\hat{\eta}_i$ is the predicted risk score, δ_i is an indicator for censoring, and T_i, T_j are the survival times.

- The numerator counts the concordant pairs, i.e., pairs where the model correctly predicts the order of events.
- The denominator normalizes by the total number of comparable pairs.

C-index by Uno (2011) with IPCW

- Uno et al. (2011) extended the C-index by introducing inverse probability of censoring weighting (IPCW)
- This method accounts for the censoring mechanism in survival data, improving the accuracy of the C-index estimation

$$\hat{\mathbb{C}}_{\text{Uno}} = \frac{\sum_{i \neq j} 1(\hat{\eta}_i > \hat{\eta}_j) 1(T_i < T_j, \delta_i = 1) \hat{w}_i(T_i)}{\sum_{i \neq j} 1(T_i < T_j, \delta_i = 1) \hat{w}_i(T_i)}, \tag{3}$$

where

$$\hat{w}_i(T_i) = \begin{cases} \frac{\delta_i}{\hat{G}(T_i)} & \text{if } T_i \leq t, \\ \frac{1}{\hat{G}(T_i)} & \text{if } T_i > t, \end{cases} \tag{4}$$

with $\hat{G}(T_i)$ is the estimated Kaplan-Meier estimate of the censoring distribution.

C-index for Specific Follow-Up Period

Time-Dependent C-index: To evaluate performance over a fixed follow-up period $[0, t^*]$, Heagerty (2005) defined the time-dependent C-index.
The time-dependent AUC at a given time t is calculated as:

$$\text{AUC}(t) = \mathbb{P}(\eta_i < \eta_j | T_i < t, T_j > t), \tag{5}$$

and the time-dependent C-index is:

$$\hat{\mathbb{C}}_{t^*} = \sum_t \widehat{\text{AUC}}(t) \cdot \text{num}(t), \tag{6}$$

where $\widehat{\text{AUC}}(t)$ is the estimated AUC at time t.

Brier Score for Survival Models

- Initially developed for weather forecasting (Brier 1950), the Brier score assesses the accuracy of probabilistic predictions.
- For binary outcomes, it is equivalent to the mean squared error.

$$\mathrm{BS} = \frac{1}{N} \sum_{i=1}^{N} \left[\hat{y}_i - y_i\right]^2 , \tag{7}$$

where $\hat{y}_i$ is the predicted probability and y_i is the actual outcome.

Brier Score for Survival with Censoring

- The Brier score is extended for survival analysis by adjusting for censoring (Graf 1999)
- The individual contributions are weighted according to the censoring distribution.

Weighted Brier Score Formula:

$$\mathrm{BS}(t) = \frac{1}{\sum_{i=1}^{N} Y_i(t)} \sum_{i=1}^{N} \hat{w}_i(t) \left[\hat{S}_i(t) - 1(T_i > t) \right]^2, \tag{8}$$

where $\hat{w}_i(t)$ is the weight for individual i, and $\hat{S}_i(t)$ is the predicted survival probability.

Integrated Brier Score (IBS)

- The Integrated Brier Score (IBS) evaluates the overall performance of a model over a period of time.
- It is defined as the integral of the Brier score over the time period $[\tau_1, \tau_2]$.

$$\text{IBS} = \frac{1}{\tau_2 - \tau_1} \int_{\tau_1}^{\tau_2} \text{BS}(t)\, dt. \tag{9}$$

- $\tau_1 = 0$ and τ_2 is typically the maximum observed follow-up time.
- The IBS provides an overall assessment of model calibration over time.

Mean Absolute Error (MAE) for Survival

- The Mean Absolute Error (MAE) measures the average absolute difference between predicted and observed survival times.
- For survival analysis, MAE is only calculated for uncensored observations.

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} \delta_i |T_i - \hat{T}_i|, \tag{10}$$

where δ_i is an indicator for event occurrence, and T_i and $\hat{T}_i$ are the observed and predicted survival times, respectively.

Addressing Censoring in MAE

Challenges with Censoring:

- A naïve approach excludes censored subjects from MAE calculation, but this may introduce bias, especially with high censoring rates.

Advanced Approaches:

- Using inverse probability censoring weighting to account for censored observations (Haider 2020)
- **MAE-margin:** Assigning a "best guess" margin time to censored subjects based on Kaplan-Meier estimators (Qin 2023)

Building prediction models

The aim of the model

- **To be considered at the very beginning:**
  - **What does the model aim to clinically achieve?**
  - **When and where will it be used?**
    - Will it be implemented in a computer or app?
- **What resources are needed?**
  - Does it rely on **readily available information**, or
  - Are **additional data** (e.g., lab values) required?
- **What is the desired "final product"?**
  - A calculator?
  - A tool that is quick and easy to use?
  - A **simplified model** for specific contexts?

Which predictors?

- **Common temptation:** Include everything
  - Risk of **overfitting**.
- **Use subject matter (clinical) expertise:**
  - Knowledge almost always exists – leverage expertise.
  - While medicine has many unknowns, there is considerable knowledge on **pathophysiology**.
  - Avoid including **irrelevant predictors**, especially when data availability is limited.
- **Consider sample size:**
  - No explicit sample size formula, but follow the rule of thumb:
    - **10 to 20 events per variable (EPV).**
  - Small EPV increases the risk of **overfitting**.

Overfitting and Underfitting

Definition: Overfitting and underfitting describe how a model generalizes from training data (Bishop 2006)

Key Concepts:

- **Overfitting:** Model fits training data too well, capturing noise.
  - Symptoms: High complexity, poor test performance.
- **Underfitting:** Model is too simple to capture data patterns.
  - Symptoms: Poor performance on both training and test data.

Goal: Achieve a balance between complexity and generalization (Hasite 2009)

Bias-Variance Trade-off

Definitions:

- **Bias:** Error due to overly simplistic assumptions.
- **Variance:** Error due to sensitivity to data fluctuations.

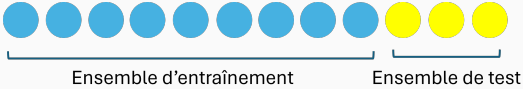
Trade-off:

- Increasing model complexity reduces bias but increases variance.
- Objective: Find the optimal balance for generalization.

Model Evaluation and Validation

Objective: Evaluate the model's ability to generalize to unseen data (Hastie 2009)

Train-test split (*e.g.*, 80%-20%)



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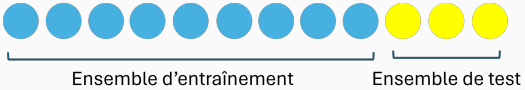
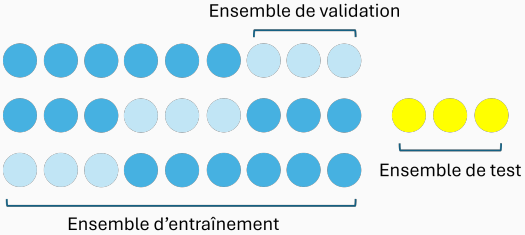


Cross-validation for robust evaluation



Hyperparameter Optimization

Definition: Hyperparameters influence model behavior and must be tuned for optimal performance.

Key Techniques:

- Grid search: Systematic search over a parameter grid.
- Random search: Randomized sampling of hyperparameter space.
- Bayesian optimization: Probabilistic model for optimization.

Warning: Avoid overfitting the validation set; use separate test data.

External validation

Rationale

We have seen before the development of the model with *internal validation*.

How well does the model perform on new unseen data?

- On another center,
- Data collected in another time batch,
- ...

Existing (yet imperfect) solutions

Challenges in External Validation:

- Data from external sources may differ in terms of **quality**, **availability**, and **measurement techniques**.
- Requires careful consideration of **heterogeneity** between training and validation datasets.

Best Practices for External Validation:

- Ensure **similarity in data structure** (e.g., outcome definitions, predictor variables).
- Perform **stratified analysis** to assess performance across different subgroups.
- Report **performance metrics** for external datasets.

Handling different data distributions (Andrew Ng's reco)

1. **Mix a small portion of external data into the training set:**
   - Take a small subset of the external validation data and combine it with the original training data.
   - This helps to expose the model to the new distribution, reducing the risk of poor performance due to distributional differences.
2. **Fine-tune the model on the augmented training set:**
   - After mixing the data, fine-tune the model on the combined training data to allow the model to adapt to the new distribution.
   - Fine-tuning ensures that the model can generalize better to both the original and the new distribution.

TP1

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